Errata and minor updates

Physical Geodesy: A Theoretical Introduction

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1. Page 3, lines 8 to 9:
   Change “as attraction, of m” to “as attraction of m”

2. Page 5. Fig. 1.3 caption:
   Change “acceleration” to “attraction”

3. Page 15. lines 1 (1.46):
   Change “series values” to “series of values”

4. Page 16, lines 2,3,5, 6 and 7:
   Change “$\hat{e}_h$” to “$\hat{e}_H$”

5. Page 16, line 7:
   Change “$\partial W/\partial h$” to “$\partial W/\partial H$”

6. Page 16, eqn (1.48):

$$g = -\frac{\partial W}{\partial H}\,.$$

7. Page 16, eqn (1.49):

$$dH = -\frac{dW}{g}\,. \tag{1}$$

8. Page 16, title of Sect. 1.3.2:
   Change to “The Geoid and Dynamic and Orthometric heights”

9. Page 20, line 2 below (1.56):
   Change “$r\psi\lambda$”to “$Or\psi\lambda$”

10. Page 22, line 2 below the section title 1.4.2:
    Change “series spherical”to “series of spherical”

11. Page 25, eq (1.81):
    Delete "dr" in the middle of the first line

12. Page 26. Below eqn (1.85) and below eqn (1.88):
    Change "disc" to "disk"

13. Page 32, 4th line of Sect. 1.5.2:
    Change "disc" to "disk"

14. Page 36, line 4:
    Change "dipole moment" to "dipole density"

15. Page 37. Line 7 from bottom:
    Change "A function satisfying" to 'A function with continuous second order partial derivatives satisfying"

16. Page 40, below eqn. (1.151):
    $(\Delta x^2 + \Delta z^2)^{1/2} = [1 + (dz/dx)^2]^{1/2}\Delta x$

17. Page 44. Line 11:
    Change "Fig. 1.17b." to "Fig. 1.17b, where the curve is an infinitesimal fraction."

18. Page 60. Line 5 from bottom:
    Change "above are all boundary" to "above are the most common boundary"

19. Page 61, add the following paragraph after the last line:
    Readers are suggested to study the uniqueness of the solutions of the boundary value problems for an internal domain, and furthermore, to show that the surface density μ in (2.55) is unique based on the uniqueness of the solutions of the first-type boundary value problems for both an external domain and an internal domain (suggestion: assume V is known on S, and prove that C does not contribute to $(\partial U/\partial n)_i$). Readers are also suggested to compare the expression of μ in (2.55) with (1.130) by setting $\hat{e}_t = \hat{e}_n$.

20. Page 89. eqn (3.45):

$$\delta = \lim_{i\to\infty}\left|\frac{C_i}{C_{i+2}}\right| = \lim_{i\to\infty}\left|\frac{(i+2)(i+1)}{i(i+1)-\alpha}\right| = 1\,.$$

21. Page 90–91, eq (3.51)–(3.55) should be:

$$\begin{aligned}\Pi > &\left[1-\frac{\alpha}{(N+1)N}\right]\cdots\left[1-\frac{\alpha}{(i-3)(i-4)}\right]\left[1-\frac{\alpha}{(i-1)(i-2)}\right]\\ &\times\left[1-\frac{\alpha}{(i+1)i}\right]\left[1-\frac{\alpha}{(i+3)(i+2)}\right]\cdots\left[1-\frac{\alpha}{(i+l+1)(i+l)}\right]\cdots\,.\end{aligned} \tag{3.51}$$

Take the logarithm on both sides

$$\ln\Pi > \ln\left[1-\frac{\alpha}{(N+1)N}\right]+\cdots+\ln\left[1-\frac{\alpha}{(i-3)(i-4)}\right] \\ +\ln\left[1-\frac{\alpha}{(i-1)(i-2)}\right]+\ln\left[1-\frac{\alpha}{(i+1)i}\right] \\ +\ln\left[1-\frac{\alpha}{(i+3)(i+2)}\right]+\cdots+\ln\left[1-\frac{\alpha}{(i+l+1)(i+l)}\right]+\cdots. \tag{3.52}$$

As

$$\lim_{l\to\infty}\frac{\alpha}{(i+l+1)(i+l)}=0\,, \tag{3.53}$$

we have

$$\lim_{i\to\infty}\left\{\ln\left[1-\frac{\alpha}{(i+l+1)(i+l)}\right]\Big/\left[-\frac{\alpha}{(i+l+1)(i+l)}\right]\right\}=1\,, \tag{3.54}$$

which can be considered as an alternative form of the formula $\lim_{x\to 0}[\ln(1+x)]/x=1$. Hence, the infinite series at the right-hand side of (3.52) converges or diverges in the same way as the series

$$\frac{\alpha}{(N+1)N}+\cdots+\frac{\alpha}{(i-3)(i-4)}+\frac{\alpha}{(i-1)(i-2)}+\frac{\alpha}{(i+1)i} \\ +\frac{\alpha}{(i+3)(i+2)}+\cdots+\frac{\alpha}{(i+l+1)(i+l)}+\cdots, \tag{3.55}$$

22. Page 91, eqn (3.58):

$$y_1^0(x)=\sum_{i=1}^{\infty}\frac{1}{2i}x^{2i} \qquad \text{and} \qquad y_2^0(x)=\sum_{i=0}^{\infty}\frac{1}{2i+1}x^{2i+1}$$

23. Page 91, add to the end of the paragraph after (3.58):
The factor M in (3.57) is dropped, since it does not affect the convergence.

24. Page 91, eqn (3.60):

$$\begin{cases} \dfrac{1}{2}[\ln(1+x)+\ln(1-x)]=-\displaystyle\sum_{i=1}^{\infty}\frac{x^{2i}}{2i}\,, \\ \dfrac{1}{2}[\ln(1+x)-\ln(1-x)]=\displaystyle\sum_{i=0}^{\infty}\frac{x^{2i+1}}{2i+1}\,. \end{cases}$$

25. Page 105, eqn (3.132):

$$P_n^k(\pm 1)=0\,, \qquad k\neq 0\,;$$

26. Page 107, eqn (3.138):
First line: Change "$\frac{n!}{(n-k)!}$" to "$\frac{n!}{(i-k)!}$".
Fifth line: Change "$(x-1)$" to "$(x-1)^n$".
Sixth line: Change "$(x+1)$" to "$(x+1)^n$".

27. Page 131, eqn (3.270):
Replace "$k = 1$" by "$k = 0$" under the summation sign.

28. Page 137, eqn (3.287)–(3.290):
Change "$\vec{r}$" to $\vec{r}'$

29. Page 136, in equation (3.285):
Change "C_1^0" to "C_1"

30. Page 142, eqn (3.314):

$$V = \frac{GM}{r} \sum_{n=0}^{\infty} \left(\frac{a}{r}\right)^n \sum_{k=0}^{n} (\bar{C}_n^k \cos k\lambda + \bar{S}_n^k \sin k\lambda) \bar{P}_n^k(\cos\theta) \,,$$

Remark: Including the degree 1 normalized coefficients is a better choice.

31. Page 144, eqn (3.321):
Replace "$k = 1$" by "$k = 0$" under the summation sign.

32. Page 144, eqn (3.322):
Replace "$k = 1$" by "$k = 0$" under the summation sign.

33. Page 145, eqn (3.327) should be:
Replace "$k = 1$" by "$k = 0$" under the summation sign.

34. Page 145, in the title of Sect. 3.6:
Replace "functions' by "function"

35. Page 145, left hand side of the second line of eqn (3.329):
Replace "a_k" by "b_k"

36. Page 147, in eqn (3.335):
Replace "2π" by "4π"

37. Page 149, line 1 below (1.347):
Change to "This formula holds for arbitrarily large N; therefore, the positive term series" and change "$\sum_{n=0}^{N}$" to "$\sum_{n=0}^{\infty}$" in (3.348)

38. Page 153, below eqn (3.370):
Change "Here we ... be written as" to "On the equator, with $\theta = \pi/2$, this formula can be written as"

39. Page 159, last sentence of the Abstract:
Change "called Geodetic Reference System (GRS)" to "which are essential for a Geodetic Reference System (GRS)"

40. Page 176, eq (4.77):
    Change "e'^2" to "e'"

41. Page 178, lines 9 to 10 from bottom:
    change "potentials in the interior are' to "potential in the interior is"

42. Page 180, lines 1 above (4.94):
    change "that $\vec{n}$ and $\vec{g}$ are parallel" to "that the directions of $\vec{n}$ and $\vec{g}$ coincide"

43. Page 186, above eqn (4.123):
    change "we have" to "we have, according to (4.80) and (1.130),"

44. Page 188, paragraph below eqn (4.135):
    Change "Borkowski (1987)" to "Borkowski (1989)"

45. Page 199, line 7 from bottom:
    Change "which are referred to as Geodetic Reference System" to "which are essential for a Geodetic Reference System"

46. Page 190, line 2 above the section title 4.6:
    Change the second "(1.172)" to "(1.173)"

47. Page 200, the line above eqn (4.180):
    Change "are the Geodetic Reference System GRS80" to "are those of the Geodetic Reference System 1980 (GRS80)"

48. Page 202, line 7 from bottom, the reference should be replaced by:
    Borkowski, K.M. (1989). Accurate algorithms to transform geocentric to geodetic coordinates. Bulletin Géodésique, 63, 50-56.

49. Page 204, the first line of the paragraph above the Section title 5.1.2:
    Change "disturbing gravity" to "gravity disturbance'

50. Page 204, the first line of the paragraph above the Section title 5.1.2:
    Change "defers" to "differs"

51. Page 209, eqn (5.20):

$$\left(\frac{\partial U}{\partial n_\Sigma}\right)_\Sigma = \left(\frac{\partial U}{\partial n_S}\right)_S + N\left(\frac{\partial^2 U}{\partial n_S^2}\right)_S = \left(\frac{\partial U}{\partial n_S}\right)_S - N\left(\frac{\partial \gamma}{\partial n_S}\right)_S .$$

52. Page 215, eqn (5.45), last term (last expression and the one before the last):
    Change "$1 + x\cos\psi$" to "$1 - x\cos\psi$".

53. Page 222, line 6 below eqn (5.69):
    Change "based on the fact that T is harmonic" to "if T is harmonic based on their spherical harmonic series"

54. Page 223, line 4:
    Change "$-g_r(R,\theta,\lambda)$" to "$-\delta g_r(R,\theta,\lambda)$"

55. Page 224, line 7:
Change “In the problems to be formulated” to “In the boundary value problems to be formulated”

56. Page 234, 2nd line below eqn (5.125):
Change “does not include its effect. The” to “does not include its effect, thus rendering the solution non-unique. The”

57. Page 246, Title of Sect. 5.5.2:
Change “Degree Power Spectrum” to “Degree Variance Power Spectrum”

58. Page 247, eqn (5.172):

$$\begin{aligned} V_n &= \frac{GM}{R}\left(\frac{R}{r}\right)^{n+1}\sum_{k=0}^{n}\left[\frac{2(2n+1)}{1+\delta_k}\frac{(n-k)!}{(n+k)!}\right]^{1/2} \\ &\quad \times(\bar{C}_n^k\cos k\lambda+\bar{S}_n^k\sin k\lambda)P_n^k(\cos\theta)\,, \end{aligned}$$

59. Page 247, eqn (5.174):

$$\begin{aligned} \frac{1}{4\pi}\int_\omega V^2 d\omega &= \left(\frac{GM}{R}\right)^2\sum_{n=0}^{n_{\max}}\left(\frac{R}{r}\right)^{2n+2}\sum_{k=0}^{n}\left[\left(\bar{C}_n^k\right)^2+\left(\bar{S}_n^k\right)^2\right] \\ &= \sum_{n=0}^{n_{\max}}\left(\frac{1}{4\pi}\int_\omega (V_n)^2 d\omega\right)\,. \end{aligned}$$

60. Page 248, Title of Fig. 5.10:
Change “Degree power spectrum variance” to “Degree variance power spectrum”

61. Page 249, Title of Sect. 5.5.3:
Change “Degree Power Spectrum” to “Degree Variance Power Spectrum”

62. Page 256, line 4 from bottom:
Delete the sentence “The correction $\Delta_1 g+\Delta_2 g$ is referred to as incomplete Bouguer correction”

63. Page 256, Last line:
Change “Bouguer correction and anomaly” to “Bouguer anomaly”

64. Page 257, lines 2-3 from bottom:
Change “Evidently, $l^2=r^2+z^2$, $\sin\beta=z/(r^2+z^2)^{1/2}$ and $dv=r d\alpha dr dz$” to “Evidently, $\sin\beta=z/l$, $l=(r^2+z^2)^{1/2}$ and $dv=r d\alpha dr dz$.”

65. Page 258, eqn (6.13):

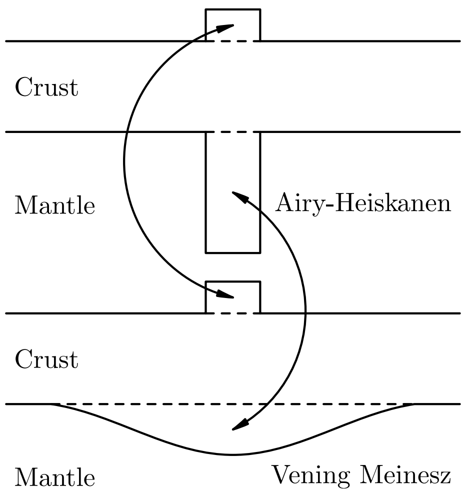
$$\Delta_3 g = G\rho\int_0^{2\pi}d\alpha\int_0^a dr\int_0^{\Delta H}\frac{zr}{(r^2+z^2)^{3/2}}dz\,,$$

66. Page 258:
Change the sentence above (6.15) to: "The gravity anomaly computed according to"

67. Page 263, eqn (6.27):

$$\Delta\rho = \rho - \rho' = -\frac{(\rho - \rho_w)P}{D - P} .$$

68. Page 265, Fig. 6.7 (It makes me feel embarrassed to have the one in the print! I went to replace it before submitting the final version to Springer, but I have mistakenly replaced it in a different draft):



69. Page 273, eqn (6.53): Add

$$\sum_{n=0}^{\infty}$$

right after the equal sign.

70. Page 275, eq (6.67)
delete the "2" in the denominator of the last term at the right-hand side

71. Page 276, eq (6.73), second line reformatted as:

$$= r^2 \int_{H_{Br}/r}^{H_{Tr}/r} \frac{(1 + 2\delta x + \delta x^2)d\delta x}{(2 - 2\cos\psi + \delta x^2)^{1/2}\left[1 + \frac{2\delta x(1 - \cos\psi)}{2 - 2\cos\psi + \delta x^2}\right]^{1/2}}$$

72. Page 277, the line above (6.78):
Change to "Define H' such that $\delta x = H'/r$. We can"

73. Page 281, eq (6.98), second line reformatted as:

$$= r \int_{H_{Br}/r}^{H_{Tr}/r} \frac{[(1 + 2\delta x + \delta x^2) - (1 + 3\delta x + 3\delta x^2 + \delta x^3)\cos\psi]\, d\delta x}{(2 - 2\cos\psi + \delta x^2)^{3/2} \left[1 + \dfrac{2\delta x(1 - \cos\psi)}{2 - 2\cos\psi + \delta x^2}\right]^{3/2}}$$

74. Page 281, the line above (6.103):
Change to "Define H' such that $\delta x = H'/r$. We can"

75. Page 285, second line of eq (6.114):
change

$$\frac{5\cos^2\Theta}{3}$$

to

$$\frac{5\cos^2\Theta}{2}$$

76. Page 289, 2nd line of eq (6.138):
Change "$4ra_r$" to "$12ra_r$" in the denominator

77. Page 289, modify the text above (6.139) to:
"The formulae of $V_{r \leq R_B}$ and $F_{r \leq R_B}$ can be obtained by simply tracking the difference between (6.116) and (6.120) for the gravitational potential and that between (6.117) and (6.121) for the gravitational attraction,"

78. Page 289, 2nd line of eq (6.140):
Change "$4ra_r$" to "$12ra_r$" in the denominator

79. Page 301, Line 5:
Change "$\sim$ 10 m for the isostatic anomaly." to "$\sim$ 10 m for the isostatic anomaly (Heiskanen and Moritz, 1967)."

80. Page 313, adjust the location of ln in eqn (6.223):

$$f_T(T, H', \psi) = \frac{R^2}{2} \Bigg\{ (x + 3\cos\psi)(1 + x^2 - 2x\cos\psi)^{1/2} + (3\cos^2\psi - 1) \times \ln\left[(1 + x^2 - 2x\cos\psi)^{1/2} + x - \cos\psi\right] \Bigg\}_{1-(T+H')/R}^{1-T/R}.$$

81. Page 319:
Add the following into the references:
Heiskanen, W.A., & Moritz,H. (1967). *Physical geodesy*. W.H. Freeman and Company.

82. Page 319, add to further reading:
Szwillus, W., Ebbing, J., & Holzrichter, N. (2016). Importance of far-field topographic and isostatic corrections for regional density modelling. *Geophysical Journal International, 207*, 274–287. https://doi.org/10.1093/gji/ggw270

83. Page 333, eq (7.39):
Add k^2 in the denominator under the second integral (same position as in the first integral).

84. Page 347, 3rd line above eqn (7.116):
Change
be readily verified using (7.24) that ... implies "$r\Delta G$ is harmonic."
to
be readily verified using (5.27) and (7.24) that $r\partial T/\partial r$ and $r\Delta G$ are harmonic.

85. Page 377, eq (7.237):
Add k^2 in the denominator under the second integral (same position as in the first integral).

86. Page 379, eq (3.246) to (3.248):
Change "$S(\psi)$" to "$K(\psi)$"

87. Page 389 line 2 to 5 below (7.285):
Change 'coastal ocean...are provided' to 'with"coastal ocean" in the statements replaced by "the neighboring regions between the regions where δg and ΔG are provided" '

88. Page 392, add the following to Further Reading:

    Guan, Z. L., & Ning, J. S. (1981). *Earth's shape and external gravity field, Vols 1* (In Chinese). Beijing, China: Press of Surveying and Mapping.

    Heiskanen, W. A., & Moritz, H. (1967). *Physical geodesy*. W.H. Freeman and Company.

89. Page 393 line 3 and page 493 line 7 from bottom (So embarrassed to misspell a colleague's name):
Change "Bgherbandi" to "Bagherbandi".

    The same correction applies to Page 493, line 7 from the bottom.

90. Page 398, line 26 (or line 9 of the third paragraph):
Change "If not, the difference between the differences" to "If not, the sum of the two differences"

91. Page 410, line 6:
Change "$n_{\max}(n_{\max}+1)$" to "$(n_{\max}+1)^2$"

92. Page 415, line 2 of the last paragraph:

    Add ", although the decrease may be oscillatory" after "β decreases with n"

93. Page 417, line 3 from bottom:
Change "$\int_{-\sigma}^{\sigma} g(x)dx = 99.73.$" to "$\int_{-3\sigma}^{3\sigma} g(x)dx = 99.73\%.$"

94. Page 423: Eqn (8.98) should be (square added)

$$\sum_{i=1}^{N}\sum_{j=1}^{2N} W_{ij}\left[\sum_{n=2}^{N-1}\sum_{k=0}^{n}(\alpha_{ijnk}\bar{c}_n^k + \beta_{ijnk}\bar{s}_n^k) - \Delta\bar{g}_{ij}\right]^2 = \min$$

95. Page 424: Change the paragraph starting the 3rd line to
"The second problem is the resolvability. In this approach, the coefficients $\bar{c}_n^k$ and $\bar{s}_n^k$ can be reliably solved only up to degree and order $N-1$ (Colombo, 1981). This could be demonstrated by examining the problem on the equator where $\theta = \pi/2$ and (8.32) degenerates to the Fourier series

$$\Delta g(\lambda) = \sum_{k=0}^{n_{\max}}(a_k \cos k\lambda + b_k \sin k\lambda)\,. \tag{8.99}$$

Hence, there are $2n_{\max}+1$ coefficients to be determined. However, there are $2N$ values of Δg on the equator. The total number of coefficients to be determined cannot be more than the total number of data. As a result, the coefficients can only be solved up to order $n_{\max} = N-1$ at most. With $n_{\max} = N-1$ as chosen in (8.98), the total number of coefficients $\bar{c}_n^k$ and $\bar{s}_n^k$ is N^2, while the total number of data $\Delta\bar{g}_{ij}$ is $2N^2$."

96. Page 424, line 4 from below:
Change "(8.98) with the upper limit of n replaced" to "(8.98) with $N-1$ replaced"

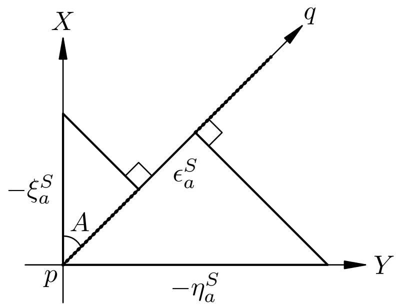
97. Page 425, line 4 from below:
Change "parameters in (8.98)" to "spherical harmonic coefficients in (8.98)"

98. Page 435, 1st paragraph:
Delete the last sentence "It is left to...is largest"

99. Page 436, add the paragraph above the title of Sect. 8.4.3:
"It is easy to realize that effect of the term $\gamma_e/\bar{\gamma}_0 = 1-\beta/3$ in (8.172) and (8.133) is very small. However, keeping it makes the numerical result 0.171 more accurate at mid-latitude, where $\sin 2B$ is largest, which is left to the readers to verify based on the exact formulae as mentioned in the previous paragraph."

100. Page 438, second line below 8.5.1:
Change "base" to "based"

101. Page 439, Fig. 8.13:



102. Page 439, change the text below (4.138) to:
"where the path from P to Q is divided into n segments through the points $P = P_0, P_1, P_2, \ldots$, and $P_n = Q$, ξ_{ai}^S and η_{ai}^S the deflection of the vertical components at P_i, A_i the azimuth of P_{i+1} as seen at P_i, and dl_i the distance between P_i and P_{i+1}."

103. Page 442 to 445, change the text below (4.154) to:
"Here the path from P to Q is also divided into n segments through the points $P = P_0, P_1, P_2, \ldots$, and $P_n = Q$, where ξ_{ai}^M and η_{ai}^M are the deflection of the vertical components at P_i, ΔG_i the gravity anomaly at P_i, A_i the azimuth of P_{i+1} as seen at P_i, dl_i the distance between P_i and P_{i+1}, and dH_i the difference of height between P_i and P_{i+1}. In the derivation, we assumed dH_i to be the difference of height determined by spirit leveling. However, as the term is small, replacing it with the normal height would not influence the accuracy of the result. If the deflection of the vertical at $P_n = Q$ is also measured, we can use a formula similar to (8.139)."

104. Page 446, add the following in Further Reading:

Guan, Z. L., & Ning, J. S. (1981). *Earth's shape and external gravity field, Vols 1* (In Chinese). Beijing, China: Press of Surveying and Mapping.

105. Page 450. eqn (9.14):
Change "$\frac{d}{\partial r}$" to "$\frac{d}{dr}$"

106. Page 452. eqn (9.19):
Change "$Y_n^k(\theta, \lambda)$" to "$Y_n(\theta, \lambda)$"

107. Page 461, the line above (9.66):
Change "We now assume" to "We now make the finer approximations"

108. Page 461, the line below (9.66):
Change "this assumption" to "these approximations"

109. Page 487. Add the following by the beginning of the line below eqn (A.57):
The substitution of (A.54) into this equation yields an equation for B, which is referred to as the latitude equation in the literature.

110. Page 492, line 2 from botom:
Change “Euler angles, $O_1x_1y_1z_1$ rotates” to ”Euler angles that $O_1x_1y_1z_1$ rotates”